

Quantitative Aptitude Master Series

Comprehensive Guide on Time & Work (Theory, 15 Solved Examples & 25 Practice MCQs)

1. Comprehensive Theory & Concepts

Introduction: Time and Work problems form a fundamental pillar of quantitative aptitude tests. The core principle revolves around the relationship between the **amount of work done**, the **time taken**, and the **rate of work (efficiency)**.

Key Fundamental Rules & Formulas:

- If a person can complete a piece of work in n days, then the work done by that person in 1 day is $1/n$.
- Conversely, if a person's 1 day's work is $1/n$, the person can finish the total work in n days.
- **Efficiency (Rate of Work):** Efficiency is inversely proportional to the time taken for the same amount of work. $\text{Efficiency} \propto 1 / \text{Time}$. If person A is twice as efficient as B, A will take half the time taken by B to complete the same work.
- **Total Work Formula:** $\text{Total Work} = \text{Efficiency} \times \text{Time Taken}$. Usually, we take Total Work as the LCM of individual times to avoid fractions.
- **M-D-H Formula (Chain Rule):**

$$\frac{M_1 \times D_1 \times H_1}{W_1} = \frac{M_2 \times D_2 \times H_2}{W_2}$$

Where M = Men/Persons, D = Days, H = Hours per day, W = Work done.

- **Working Together:** If A can do a piece of work in x days and B in y days, together they can finish the work in $\frac{x \times y}{x + y}$ days.

2. Solved Examples (15 Detailed Solutions)

Example 1: Basic Individual Work

A can do a piece of work in 10 days and B can do it in 15 days. How long will they take together to finish the work?

Solution:

1 day work of A = $1/10$, 1 day work of B = $1/15$.

(A + B)'s 1 day work = $1/10 + 1/15 = (3 + 2)/30 = 5/30 = 1/6$.

Therefore, both together will finish the work in **6 days** (or using LCM method: Total work = LCM(10,15) = 30 units. Efficiencies: A=3, B=2. Combined = 5. Time = $30/5 = 6$ days).

Example 2: One person leaving before completion

A and B can complete a work in 12 and 18 days respectively. They work together for 4 days, then A leaves. In how many days will B finish the remaining work?

Solution:

Let Total Work = LCM(12, 18) = 36 units. Efficiency of A = 3 units/day, B = 2 units/day.

Work done by (A + B) in 4 days = $4 \times (3 + 2) = 20$ units.

Remaining work = $36 - 20 = 16$ units.

Time taken by B to finish 16 units = $16 / 2 = 8$ days.

Example 3: Efficiency Comparison

A is 3 times as fast as B and takes 60 days less than B to complete a work. Find the time taken by both working together.

Solution:

Ratio of efficiencies (A : B) = 3 : 1. Ratio of time taken = 1 : 3.

Difference in time ratios = $3 - 1 = 2$ units, which equals 60 days. So 1 unit = 30 days.

Time taken by A = 30 days, B = 90 days. LCM = 90 units.

Combined efficiency = $3 + 1 = 4$ units/day. Time = $90 / 4 = 22.5$ days.

Example 4: Pipe and Cistern / Men and Days

If 12 men can complete a work in 25 days, in how many days will 15 men complete the same work?

Solution:

Using $M_1 \times D_1 = M_2 \times D_2$

$$12 \times 25 = 15 \times D_2$$

$$D_2 = (12 \times 25) / 15 = 300 / 15 = \mathbf{20 \text{ days.}}$$

Example 5: Working on Alternate Days

A and B can do a piece of work in 20 and 30 days respectively. If they work on alternate days starting with A, in how many days will the work be finished?

Solution:

Total Work = LCM(20, 30) = 60 units. Efficiency of A = 3, B = 2.

Work done in 2 days (A then B) = 3 + 2 = 5 units.

In 12 cycles ($12 \times 2 = 24$ days), work done = $12 \times 5 = 60$ units.

Thus, the work is finished in exactly **24 days**.

Example 6: Men, Women and Children

If 2 men or 3 women can do a piece of work in 40 days, find the time taken by 4 men and 5 women to complete the same work.

Solution:

$$2 \text{ Men} = 3 \text{ Women} \Rightarrow 1 \text{ Man} = (3/2) \text{ Women.}$$

$$\text{Total Work} = 2 \text{ Men} \times 40 \text{ days} = 2 \times (3/2) \text{ Women} \times 40 = 120 \text{ women-days.}$$

$$\text{Now convert 4 Men into women: } 4 \text{ Men} = 4 \times (3/2) = 6 \text{ women.}$$

$$\text{Total workforce} = 6 \text{ women} + 5 \text{ women} = 11 \text{ women.}$$

$$\text{Time} = 120 / 11 = \mathbf{10\frac{10}{11} \text{ days.}}$$

Example 7: Joining after some days

A can do a work in 20 days and B in 40 days. They start together, but A leaves 5 days before the completion of the work. In how many days was the total work finished?

Solution:

LCM = 40 units. Efficiency: A = 2, B = 1.

Since A left 5 days before, in those last 5 days only B worked. Work done by B in 5 days = $5 \times 1 = 5$ units.

Remaining work before that = $40 - 5 = 35$ units, done by (A + B) together.

Time taken for 35 units = $35 / (2 + 1) = 35/3$ days.

Total time = $35/3 + 5 = 50/3 = 16\frac{2}{3}$ days.

Example 8: Chain Rule with Hours

If 15 men working 8 hours a day can reap a field in 20 days, in how many days will 20 men working 6 hours a day reap the same field?

Solution:

Formula: $M_1 D_1 H_1 = M_2 D_2 H_2$

$$15 \times 20 \times 8 = 20 \times D_2 \times 6$$

$$2400 = 120 \times D_2 \Rightarrow D_2 = 2400 / 120 = 20 \text{ days.}$$

Example 9: Fractional Work Completion

A completes $\frac{1}{4}$ of a work in 10 days, B completes 40% of the same work in 15 days, and C completes $\frac{1}{3}$ of the work in 13 days. Who is the most efficient?

Solution:

Total time for A = $10 \times 4 = 40$ days.

Total time for B = $15 / 0.4 = 37.5$ days.

Total time for C = $13 \times 3 = 39$ days.

Since B takes the least time (37.5 days), **B is the most efficient.**

Example 10: Three Persons Together

A, B, and C can complete a piece of work in 12, 15, and 20 days respectively. Working together, how long will they take to complete it?

Solution:

LCM = 60 units. Efficiencies: A = 5, B = 4, C = 3.

Combined efficiency = $5 + 4 + 3 = 12$ units/day.

Time = $60 / 12 = 5$ days.

Example 11: A and B, B and C, C and A combinations

A and B can do a work in 12 days, B and C in 15 days, and C and A in 20 days. In how many days can A alone finish the work?

Solution:

LCM = 60 units. (A+B)'s eff = 5, (B+C)'s = 4, (C+A)'s = 3.

Sum of 2(A+B+C) efficiencies = $5 + 4 + 3 = 12$.

(A+B+C)'s efficiency = $12 / 2 = 6$.

A's efficiency = (A+B+C) - (B+C) = $6 - 4 = 2$ units/day.

Time taken by A alone = $60 / 2 = 30$ days.

Example 12: Leaving and Joining Sequence

A and B can do a work in 18 and 24 days. They work together for 8 days, then B leaves. How long will A take to finish the remaining work?

Solution:

LCM = 72 units. Efficiencies: A = 4, B = 3.

Work done in 8 days together = $8 \times (4 + 3) = 56$ units.

Remaining work = $72 - 56 = 16$ units.

Time taken by A = $16 / 4 = 4$ days.

Example 13: Wages in Proportion to Efficiency

A and B undertake a project for Rs. 4500. A can do it in 10 days and B in 15 days. With C's help, they finish it in 5 days. What is C's share?

Solution:

LCM = 30 units. Efficiencies: A = 3, B = 2.

(A + B + C)'s combined efficiency = $30 / 5 = 6$.

C's efficiency = $6 - (3 + 2) = 1$ unit/day.

Wages are distributed in ratio of efficiencies (3 : 2 : 1).

C's share = $(1 / 6) \times 4500 = \text{Rs. 750}$.

Example 14: Percentage Efficiency Increase

P is 40% more efficient than Q. If Q takes 21 days to complete a work, how many days will P take?

Solution:

Efficiency ratio (P : Q) = 140 : 100 = 7 : 5.

Time ratio (P : Q) = 5 : 7.

If Q's time = 7 units = 21 days (1 unit = 3 days), then P's time = 5 units = $5 \times 3 = 15$ days.

Example 15: Engine / Machine Output Work

3 engines working 8 hours a day consume 6 liters of diesel in 10 days. How many liters will 4 engines working 6 hours a day consume in 15 days?

Solution:

Using Chain Rule: $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$

$\frac{3 \times 10 \times 8}{6} = \frac{4 \times 15 \times 6}{W_2}$

$\frac{240}{6} = \frac{360}{W_2} \Rightarrow 40 = 360 / W_2 \Rightarrow W_2 = 360 / 40 = 9$ liters.

3. Practice Questions (25 Multiple Choice Questions)

Q1. A can complete a work in 12 days and B in 20 days. If they work together, in how many days will they finish the work?

A) 6 days

B) 7.5 days

C) 8 days

D) 9 days

Q2. If 6 men can do a piece of work in 10 days, how many men are needed to complete the same work in 5 days?

- | | |
|-----------|-----------|
| A) 8 men | B) 10 men |
| C) 12 men | D) 15 men |

Q3. A is twice as good a workman as B and together they finish a piece of work in 14 days. In how many days can A alone finish the work?

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|------------|------------|
| A) 21 days | B) 28 days |
| C) 42 days | D) 20 days |

Q4. A, B, and C can finish a work in 10, 12, and 15 days respectively. Working together, how long will they take?

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|-----------|-----------|
| A) 3 days | B) 4 days |
| C) 5 days | D) 6 days |

Q5. P and Q can do a job in 10 days, Q and R in 15 days, and R and P in 20 days. How long will R take alone?

- | | |
|------------|-------------|
| A) 80 days | B) 120 days |
| C) 60 days | D) 90 days |

Q6. 10 women can complete a project in 15 days and 15 children can complete it in 20 days. How many days will 5 women and 10 children take together?

- | | |
|------------|------------|
| A) 12 days | B) 15 days |
| C) 10 days | D) 18 days |

Q7. A can do a work in 24 days. If B is 50% more efficient than A, how many days will B take to do the same work?

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|------------|------------|
| A) 12 days | B) 16 days |
| C) 18 days | D) 20 days |

Q8. A and B start working together, but A leaves after 3 days. If A takes 15 days and B takes 10 days individually, total days to finish the work:

- | | |
|-----------|------------|
| A) 7 days | B) 8 days |
| C) 9 days | D) 10 days |

Q9. If 12 men or 18 women can reap a field in 14 days, in how many days can 8 men and 16 women reap the same field?

- | | |
|------------|------------|
| A) 9 days | B) 8 days |
| C) 10 days | D) 12 days |

Q10. A pipe can fill a tank in 5 hours, but due to a leak it takes 6 hours. In how much time can the leak empty the full tank?

- | | |
|-------------|-------------|
| A) 25 hours | B) 30 hours |
| C) 35 hours | D) 40 hours |

Q11. A does $\frac{4}{5}$ of a work in 20 days. He then calls B and they finish the remaining work in 3 days. How long would B alone take?

- | | |
|--------------|------------|
| A) 37.5 days | B) 30 days |
| C) 45 days | D) 25 days |

Q12. X can do a piece of work in 40 days. He works for 8 days and then Y finishes the remaining work in 16 days. Together they would take:

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|------------|---------------|
| A) 12 days | B) 15 days |
| C) 10 days | D) 13.33 days |

Q13. 20 men working 5 hours a day can complete a work in 10 days. How many hours a day should 25 men work to finish it in 8 days?

- | | |
|------------|------------|
| A) 4 hours | B) 5 hours |
| C) 6 hours | D) 8 hours |

Q14. A and B working together can finish a job in 6 days. A alone can finish it in 10 days. How long will B alone take?

- | | |
|------------|------------|
| A) 12 days | B) 15 days |
| C) 18 days | D) 20 days |

Q15. Three men A, B, C complete a job. A works for 2 days, B for 3 days, and C for 4 days. Their daily wages are in ratio 5:6:4, and total earnings is Rs. 1100. B's share is:

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|------------|------------|
| A) Rs. 350 | B) Rs. 450 |
| C) Rs. 400 | D) Rs. 500 |

Q16. A is 3 times as efficient as B, and hence takes 40 days less than B to finish a work. Time taken when working together:

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|------------|------------|
| A) 12 days | B) 15 days |
| C) 10 days | D) 20 days |

Q17. If 5 engines consume 9 metric tonnes of coal when each operates 9 hours a day, how many metric tonnes will be consumed by 8 engines operating 10 hours a day?

- | | |
|--------------|--------------|
| A) 14 tonnes | B) 16 tonnes |
| C) 18 tonnes | D) 20 tonnes |

Q18. A can do a work in 10 days, B in 12 days, and C in 15 days. They all start together, but A leaves 2 days before completion. Total days:

- | | |
|-----------|-------------|
| A) 4 days | B) 5 days |
| C) 6 days | D) 4.5 days |

Q19. Two workers A and B can build a wall in 30 and 36 days respectively. If they work together with a third worker C, they build it in 12 days. C alone can build it in:

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|------------|------------|
| A) 30 days | B) 40 days |
| C) 45 days | D) 60 days |

Q20. A contractor undertook to finish a road in 40 days and employed 100 men. After 24 days, only half the work was done. How many extra men should he employ now?

- | | |
|-----------|-----------|
| A) 50 men | B) 60 men |
| C) 40 men | D) 30 men |

Q21. A, B can do a work in 16 days, B, C in 24 days, C, A in 48 days. In how many days can they complete it together?

- | | |
|---------------|------------|
| A) 12 days | B) 15 days |
| C) 21.33 days | D) 18 days |

Q22. A is thrice as efficient as B and takes 20 days less than B to finish a task. Find time taken by B alone.

- | | |
|------------|------------|
| A) 30 days | B) 40 days |
| C) 45 days | D) 60 days |

Q23. A container is filled by tap A in 6 mins and emptied by tap B in 8 mins. If both are opened simultaneously, when will the container fill?

- | | |
|------------|------------|
| A) 20 mins | B) 24 mins |
| C) 18 mins | D) 30 mins |

Q24. 4 men and 6 women can complete a task in 8 days, while 3 men and 7 women can do it in 10 days. In how many days can 10 women complete it?

- | | |
|------------|------------|
| A) 40 days | B) 50 days |
| C) 60 days | D) 45 days |

Q25. A does half as much work as B in three-fourths of the time. If together they take 18 days to complete a work, how much time does B take alone?

- | | |
|------------|------------|
| A) 30 days | B) 40 days |
| C) 50 days | D) 45 days |

Answer Key (Practice Questions)

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
1	B (7.5 days)	8	A (7 days)	15	B (Rs. 450)	22	A (30 days)
2	C (12 men)	9	A (9 days)	16	A (12 days)	23	B (24 mins)
3	A (21 days)	10	B (30 hours)	17	B (16 tonnes)	24	A (40 days)
4	B (4 days)	11	A (37.5 days)	18	B (5 days)	25	B (40 days)
5	C (60 days)	12	C (10 days)	19	C (45 days)		
6	C (10 days)	13	B (5 hours)	20	A (50 men)		
7	B (16 days)	14	B (15 days)	21	A (12 days)		